DATA ANALYTICS FOR BUSINESS

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Censored and Truncated Models

Learning Objectives

- Articulate the differences among
  - Corner solution response
  - Censored data
  - Truncated data
- Tobit Model
  - Describe the setup of Tobit models
  - Explain how to estimate Tobit models
  - Interpret the partial effects from Tobit models
- Truncated Model:
  - Explain how to estimate truncated models
  - Interpret the partial effects from truncated models

Corner Solution Response

- Limited Dependent Variable Models:
  - Binary Response
  - Corner Solution Response
  - Censored Data
  - Truncated Data
  - Count Variable

  } Tobit Model

- Corner Solution Response:
  - Dependent variable's value is restricted to a certain range;
  - Values within the valid range are continuous;
  - Values outside the valid range take the "corner solution";
  - Corner solution value accounts for a significant portion in the data
  - Examples:
    - Mobile data consumption volume: ≥ 0
    - Charitable contribution amount: ≥ 0
    - Share of wallet: $[0,1]$

Tobit Model

- Tobit Model:

$$y^* = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u \qquad u \sim N(0, \sigma^2)$$

$$y = \begin{cases} y^* & \text{if } y^* > 0 \\ 0 & \text{if } y^* \leq 0 \end{cases}$$

  - σ^2: variance of the error term, to be estimated together with β
    identifiable because of the continuous values of y (unlike Logit/Probit)
  - The threshold has been normalized to zero, but can be easily adjusted
  - Model can be easily adjusted to allow the corner solution at the lower bound and/or the upper bound

Estimating Tobit Model

- Deriving the likelihood function:

$$\text{For } y > 0,\ \ f(y|x) = f(u = y - x\beta) = \frac{1}{\sigma}\phi\left(\frac{y - x\beta}{\sigma}\right),$$

$$\text{For } y = 0,\ P(y = 0|x) = P(y^* \leq 0|x) = P(u \leq -x\beta)$$
$$= P\left(\frac{u}{\sigma} \leq -\frac{x\beta}{\sigma}\right) = \Phi\left(-\frac{x\beta}{\sigma}\right) = 1 - \Phi\left(\frac{x\beta}{\sigma}\right)$$

  - $\Phi(\cdot)$ *cdf*, $\phi(\cdot)$ *pdf*: standard normal distribution $N(0,1)$

- Likelihood function:

$$\mathcal{L}(\beta, \sigma) = \prod_{y_i > 0} [\sigma^{-1}\phi((y_i - x_i\beta)/\sigma)] \prod_{y_i = 0} [1 - \Phi(x_i\beta/\sigma)]$$

- Tobit model can be estimated by **maximum likelihood estimation (MLE)**

$$(\hat{\beta}, \hat{\sigma}) = \underset{\beta,\sigma}{\operatorname{argmax}}\, \mathcal{L}(\beta, \sigma)$$

Partial Effects

- Expected value of y

$$E(y|x) = P(y > 0|x) \cdot E(y|y > 0, x) + P(y = 0|x) \cdot E(y|y = 0, x)$$
$$= P(y > 0|x) \cdot E(y|y > 0, x)$$
$$= \Phi\left(\frac{x\beta}{\sigma}\right) x\beta + \sigma\phi\left(\frac{x\beta}{\sigma}\right)$$

- Partial (marginal) Effects

$$\boldsymbol{\frac{\partial E(y|x)}{\partial x_j}} = \frac{\partial P(y > 0|x)}{\partial x_j} \cdot E(y|y > 0, x) + P(y > 0|x) \cdot \frac{\partial E(y|y>0,x)}{\partial x_j}$$
$$= \boldsymbol{\beta_j \Phi\left(\frac{x\beta}{\sigma}\right)}$$

  - Two parts of effects:
    - $\frac{\partial P(y > 0|x)}{\partial x_j}$: as x_j changes, individuals originally with $y = 0$ may choose $y > 0$
    - $\frac{\partial E(y|y>0,x)}{\partial x_j}$: x_j may directly affect the y value for those with $y > 0$ already

Limitations of Tobit Model

- Limitations of Tobit:
  - The expected value of y conditional on $y > 0$, $E(y|y > 0)$, is closely linked to the probability of $y > 0$, $P(y > 0|x)$
  - $\frac{\partial P(y > 0|x)}{\partial x_j}$ and $\frac{\partial E(y|y>0,x)}{\partial x_j}$ may not be in the same direction, but Tobit effectively averages these two effects
  - More general: two-part model

Censored Data

- Censored Data vs. Corner Solution Response
  - Very similar, often mixed together and use interchangeably
  - Corner solution response:
    - Censoring is due to economic constraints
    - Observed y (including the “corner solution”) is what is meaningful
  - Censored data:
    - Censoring is due to limited data observability
    - Underlying (uncensored) y^* is what is meaningful and of interest

- Censored Data
  - Still a random sample from the population
  - All x variables are observed
  - Some y are not observable above/below certain thresholds
    - But we still know whether y is above or below the threshold
    - Often due to survey design (e.g., top/bottom coding, “Income ≥ 10,000”, “Net worth ≤ 0”)

Censored Data Model

$$y^* = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u \qquad u \sim N(0, \sigma^2)$$

$$y = \begin{cases} c_1 & \text{if } y^* > c_1 \\ y^* & \text{if } c_2 \leq y^* \leq c_1 \\ c_2 & \text{if } y^* < c_2 \end{cases}$$

- Estimation
  - Can be estimated just like Tobit model, using MLE
- Interpretation
  - Interpreting the results is straightforward because we are interested in y^*
  - $E(y^*|x) = x\beta$
  - Partial effect simply equals the coefficient estimate, $\frac{\partial E(y^*|x)}{\partial x_j} = \beta_j$, just like the linear regression model!

Truncated Data

- Truncated vs. Censored Data
  - Data Censoring:
    - Still a random sample, always observe explanatory variable x for all individuals
    - Limited information on y if it is beyond the threshold (still know it is greater/less than the threshold)
  - Data Truncation:
    - A subset of the population is completely missing, no longer a random sample
    - If y is beyond the threshold, do not observe either x or y --- no information at all!

- Truncated Data
  - We do not observe a subset of the population based on the value of y
  - If y is above/below certain thresholds, both the *dependent* variable y and the ***independent*** variables x are NOT observed at all
  - No longer a random sample!
  - Due to data observability and/or sample selection issues

Truncated Data Model

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + u \qquad u \sim N(0, \sigma^2)$$

(x, y) is missing if $y > c$

- Likelihood function

$$g(y|x, c) = \frac{f(y - x\beta)}{F(c - x\beta)}$$

  - $f(\cdot)$ pdf, $F(\cdot)$ cdf: normal distribution $N(0, \sigma^2)$
  - *Conditional* on the fact that $y \leq c$ (i.e., normalize by $F(c - x\beta)$)
  - Can be estimated by MLE

- Interpretation
  - Interpretation is straightforward, just like the linear model
  - Partial effect $\frac{\partial E(y|x)}{\partial x_j} = \beta_j$ ($\hat{\beta}$ is estimated by the truncated model, corrected for the sample selection bias

R Functions

- **censReg** package: for Tobit model
  - **censReg**(formula, left=0, right=Inf, data): Estimate a Tobit model
  - **margEff**(*Tobit.result*, xValues): compute the marginal effects for a Tobit model
    - Notice that xValues can only be a vector (rather than a matrix) – only applied to one point at a time
    - Without xValues, evaluated at the mean of the explanatory variables in the sample

- **truncreg** package: for Truncated regression
  - **truncreg**(formula, data, point=0, direction="left")

Summary

- Differences among
  - Corner solution models
  - Censored models
  - Truncated models
- Tobit Model
  - Setup of Tobit models
  - How to estimate Tobit models
  - Partial effects from Tobit models
- Truncated Model:
  - How to estimate truncated models
  - Partial effects from truncated models